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A Model for Integrating Artificial Intelligence with ERP Systems - Towards Autonomous Business Management Systems

Damian Dziembek^a, Tomasz Turek^{a*}

^a*Czestochowa University of Technology, Dabrowskiego 69, Czestochowa 42-201, Poland*

Abstract

Contemporary ERP (Enterprise Resource Planning) systems play a central role in business management in the era of digital transformation. The development of artificial intelligence (AI) technology significantly influences the architecture and functionality of ERP systems, transforming them into intelligent decision-support platforms. The scope and depth of AI integration with ERP may vary, which is why the article identifies three main areas of support: process automation, data analytics and decision support, as well as user interface personalization and interactivity. Within each of these areas, four levels of increasingly advanced AI applications in ERP systems are distinguished. The proposed model also illustrates the evolution of ERP systems—from reactive to proactive solutions capable of metacognition, performance optimization, and autonomous decision-making. As a result, adaptive, cognitive, and autonomous ERP systems are distinguished. The article is based on a review of literature in the fields of economic informatics and management, supplemented by selected business case studies. It is conceptual and analytical in nature and serves as a starting point for further research and the planning of AI implementations in ERP environments.

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1. Introduction

Nowadays, effective and efficient support for business management, regardless of the size and industry of the company, requires the use of various types of IT systems, among which ERP (Enterprise Resource Planning) systems are of key importance. As a central element of an enterprise's information infrastructure, ERP systems integrate

* Corresponding author. Tel.: +48 34 3250391; fax: +48 34 3250391

E-mail address: tomasz.turek@pcz.pl

resources and business processes within a single coherent platform. Since their inception as tools for production resource planning, ERP systems have undergone significant technological and functional evolution. In the era of rapid development of information and communication technologies, the integration of ERP with artificial intelligence (AI) tools is becoming increasingly important. The use of AI, including machine learning algorithms, natural language processing, neural networks, and computer vision – significantly expands the capabilities of ERP systems: from task and process automation, real-time data analysis, and forecasting, to knowledge-based decision support and profound changes in user interaction. This integration can vary in scope from simple support mechanisms to intelligent, self-learning systems autonomously managing business processes. AI thus influences not only how ERP systems operate, but also how entire enterprises function.

The scale and complexity of these changes call for a systematic classification of the levels of AI integration with ERP, as well as an assessment of their impact on organizational effectiveness. Previous studies have mainly focused on selected AI applications within specific ERP modules, while there is a lack of a holistic approach that would allow for classification of the various types and levels of AI integration with ERP systems, and thereby a deeper understanding of the scale and depth of this transformation. The aim of this article is to present a model of the levels of AI application in ERP systems. This model is significant both scientifically as a contribution to understanding the maturity and evolution of ERP in the context of digital transformation and practically, by supporting managers and providers in planning AI-based implementations.

2. The evolution of ERP systems

ERP systems are currently widely used in various types of enterprises, enabling effective resource management, fast information flow, efficient and effective business process execution, and real-time decision making. In the literature, ERP is defined as configurable software packages that integrate information and processes across an organization in real time using a single, shared database or as a software system designed to facilitate the automation and management of essential business operations within enterprises in order to achieve optimal performance [1].

ERP systems evolved from early business IT solutions of the 1960s, through MRP and MRP II in the 1970s–1980s, into comprehensive platforms integrating core business functions. At the turn of the 21st century, ERP II extended system functionalities beyond organizational boundaries, integrating with CRM, SCM, and e-commerce. This shift was accompanied by a move toward modular, service-oriented architectures (SOA), enhancing system flexibility. The emergence of ERP III brought mobile access, customer participation in processes, and the adoption of Grid and Cloud Computing. Since 2010, cloud-based ERP systems using SaaS models have enabled flexible, scalable, and cost-effective solutions, accessible via mobile and web platforms.

The modern generation of ERP (often referred to as postmodern ERP) is based on cloud computing, microservices, open APIs, and integration with Industry 4.0 technologies such as the Internet of Things (IoT), big data analytics, and artificial intelligence (AI). In this context, ERP is no longer just an operational tool, but a platform for digital transformation, enabling intelligent, adaptive, and predictive management of an organization. At the same time, the role of ERP in the organizational structure is changing - instead of merely mapping existing processes, these systems are increasingly shaping new operating models and supporting innovation. Reduced data processing costs, the development of cloud services (SaaS), and growing computing power are enabling the implementation of ERP not only in large corporations but also in the SME sector, contributing to the democratization of advanced IT solutions. In the context of integration with AI, modern ERP systems are increasingly equipped with intelligent decision support, process automation (e.g., RPA), natural language processing (e.g., chatbots for customer service), and predictive operational analytics. This blurs the boundaries between ERP as a transactional system and analytical and cognitive systems, which naturally leads to the need to classify the levels of AI advancement in ERP, which is the subject of the rest of this article.

3. Levels of artificial intelligence adaptation in ERP systems

Currently, ERP systems are not only a management support tool, but also a key component of the digital transformation of modern enterprises. The need to meet the requirements of the modern market environment (e.g., increasing complexity of business processes, rapid response to market changes) means that ERP systems are

undergoing significant changes. One of the important factors that has created new opportunities for the development of ERP systems is the dynamic development of artificial intelligence (AI) technology. In general, AI, understood as the ability of computer systems to perform tasks requiring human intelligence, such as reasoning, learning, planning, pattern recognition, and natural language processing, extended by the ability to make decisions and adapt to situations [2], provides new opportunities to increase the value of ERP systems. The application of AI technologies such as machine learning (ML), deep learning (DL), natural language processing (NLP), intelligent process automation (IPA), predictive analytics, and computer vision (CV) in ERP systems enables task automation, large data set analysis, and the provision of forecasts to support decision-making, improve operational efficiency, reduce costs, optimize resource management, forecast demand, and improve customer service [3] [4]. In the context of ERP, AI can be used for process automation, data analysis, forecasting, inventory management, personalization, and real-time decision making. AI is changing the way ERP systems work by enabling them to learn dynamically, make predictive decisions, and interact with users in a contextual and personalized way. As a result, the potential of artificial intelligence enables the transformation of ERP systems into intelligent enterprise management support systems that can significantly improve the efficiency, effectiveness, and effectiveness of business processes, reduce operating costs, increase the quality of decisions made, and improve cooperation and satisfaction of customers, partners, and employees. Thus, intelligent ERP systems enable companies to better manage their resources, predict market trends faster and more effectively, and respond quickly to changing market conditions.

The scope and depth of AI tool application in ERP systems can vary. Figure 1 highlights three main areas in which AI supports ERP systems: Process Automation, Data Analytics and Decision Support, and Interactivity, Multimedia, and Interface Personalization. Within each of these areas, four levels of AI application in ERP systems are distinguished. In the proposed framework, ERP systems are based on multilayered integration of AI technologies, which is reflected in their functional and structural architecture. The presented model of ERP system evolution illustrates their gradual transformation - from reactive systems to proactive systems, and ultimately to autonomous systems, characterized by metacognitive capabilities, optimization of internal processes, and the ability to make independent decisions without human intervention. The model of AI application levels in ERP systems was developed using an analytical approach based on a review of the literature in the field of economic informatics and management, supplemented by real-world business examples.

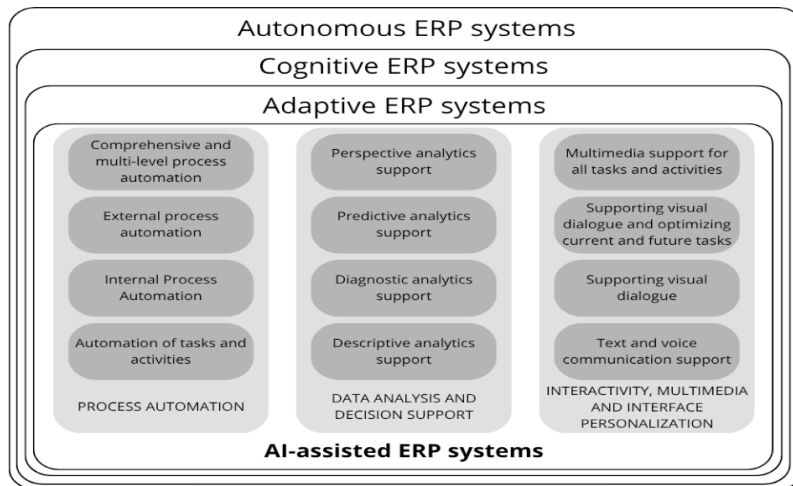


Fig. 1. Main areas of ERP system supported by AI

Within the **first area** of ERP system support by AI tools, i.e. **Process Automation**, the following levels have been distinguished:

- **Automation of daily tasks and activities** - at the lowest level in ERP systems, routine, repetitive and low-complexity activities and tasks are automated using RPA (Robotic Process Automation) and simple AI models.

These automatically performed tasks in ERP systems include entering data from invoices and financial documents, verifying documents, generating orders based on predefined rules, automatic posting of operations, and generating reports and statements. At this level, it is also possible to use virtual voice assistants or chatbots that handle simple user queries regarding frequently performed tasks (e.g., order status checks, invoice approvals, etc.). Automation at this level improves operational efficiency by reducing the number of manual and labor-intensive tasks and limiting human error [5][6].

- At the second level of AI integration with ERP systems, tasks within a single process or group of related **internal processes** are automated. AI-supported ERP systems take over repetitive and routine tasks, but human involvement is not completely eliminated. Staff receive recommendations on how to proceed and use them to make management decisions. These systems, which use machine learning and analysis of historical and current data, are able to suggest optimal actions and dynamically adjust business processes. The use of advanced ML algorithms, process mining, and decision agents enables the optimization of key areas of a company's operations, such as logistics, production, human resource management, and finance. Automation covers not only individual activities, but increasingly also complex sequences of activities, such as order-to-cash or purchase-to-pay processes. In this context, a multi-agent approach is becoming particularly important, in which autonomous AI components work together to execute processes while monitoring multiple interdependent parameters and informing employees of the need for adjustments. Automation at this level translates into increased operational efficiency, shorter process execution times, lower costs, and improved quality, among other things, by eliminating human error - which in turn contributes to increased customer satisfaction [7][8].
- **External process automation** - means extending automation beyond the boundaries of the organization to include cooperation with business partners - suppliers, customers, and institutions. At this level, repetitive activities carried out jointly with partners are automated, which reduces the involvement of employees on both sides. The human factor is not completely eliminated - decisions on the scope of automation are made jointly by the participants in the processes. Key areas covered by automation include logistics, commercial, and financial processes - for example, automated invoice circulation in the purchase-payment cycle. AI-assisted ERP systems can integrate data from various sources (SCM, CRM, purchasing platforms) to predict external events such as delivery delays, changes in customer demand or partner risks. This enables better coordination of activities and optimization of inventory and goods flows. Advanced AI technologies such as machine learning, process mining, chatbots, decision agents, and multi-agent systems are used to automate these processes. The latter play a particularly important role in the integration of processes across different organizations - they automatically process data (e.g., information about inventory, orders, invoices) and notify participants when action needs to be taken. The benefits of this automation include greater transparency and trust between partners, reduced delays, better communication and data synchronization, which translates into improved operational performance, such as shorter order fulfillment times. The elimination of manual operations and errors in document exchange also contributes to financial benefits [9].
- **Comprehensive and multi-level process automation** - at the highest level, an ERP system with AI tools becomes the central element of a digital ecosystem in which automated processes connect multiple organizations at different levels. Multi-Agent Systems are used here, where autonomous AI agents collaborate to manage selected sub-processes or processes carried out both within the company (e.g. trade, supply chain management, finance, HR) and in cooperation with partners, customers, and institutions (e.g., the entire supply chain from raw material ordering to final distribution, as well as performing many tasks simultaneously, taking into account the operational goals of individual departments or cooperating organizations). As a result, at this level, there may be dynamic adjustments to prices and terms of cooperation with partners, changes in logistics routes and optimization of inventories at various partner locations in real time, as well as production planning based on market data. At this level, the human factor plays mainly a monitoring and decision-making role, leaving the majority of repetitive tasks and activities in all processes to ERP systems and AI tools. The business effects include increased supply chain flexibility, the ability to create new cooperation models, and an improved market position for cooperating companies [10][4][11].

A synthetic overview of AI support levels in ERP systems in the context of process automation is presented in Table 1.

Table 1. AI support levels in ERP systems in the context of process automation

Level	Sample tasks	AI technologies used	Examples of solutions/effects
Automation of daily tasks and activities	automatic invoice and document posting (OCR + RPA), simple queries and customer request handling via chatbots (AI NLP); generation of standard purchase orders to key suppliers based on order history	RPA (bots executing scripts in ERP systems), NLP models (voice and chat assistants), OCR, and simple ML algorithms for document classification (invoice field recognition and filling).	Comarch ERP (RPA and chatbot functionalities), Oracle Fusion Cloud ERP (AI Digital Assistant), Microsoft Dynamics 365 with Copilot module, SAP Business AI (Joule).
Internal process automation	automation of tasks within a single process or a group of related internal processes; the ability to support more complex sequences of activities, such as <i>order-to-cash</i> or <i>purchase-to-pay</i> processes.	RPA, NLP models, ML, process mining and decision agents	Microsoft Dynamics: financial and sales process automation; SAP Intelligent Robotic Process Automation (RPA)
External process automation	extending automation beyond the boundaries of the organization to include collaboration with business partners - suppliers, customers, and institutions; automation of repetitive tasks performed with partners (e.g., logistics, commercial, and financial processes)	ML (machine learning), process mining, chatbots, decision agents, and multi-agent systems.	IFS.ai: automatic order and report generation, invoice and receipt content recognition; Microsoft Dynamics: financial and sales process automation; SAP Intelligent Robotic Process Automation (RPA)
Comprehensive and multi-levels process automation	vertical and horizontal support for internal and external processes; ERP with AI tools is becoming a central element of the digital ecosystem, where automated; processes connect multiple organizations at different levels;	Multi-Agent Systems; decision agents; NLP models, ML, process mining and decision agents	IFS.ai Copilot: autonomous recommendations and resource optimization; Microsoft Dynamics 365 Copilot: generative AI support; SAP Business Technology Platform: adaptive business processes and AI-driven workflows

Within **the second area** of ERP system support through AI tools, i.e. **Data Analytics and Decision Support**, the following levels were distinguished:

- **Descriptive analytics support** - which focuses on the systematic collection, aggregation, and visualization of historical data to create a complete picture of past operational and financial events. In AI-assisted ERP environments, this analytics goes beyond traditional reporting, enabling dynamic dashboards, real-time analysis, and recognition of user or customer behavior patterns. Through the use of data mining methods, these systems are able to not only present data, but also detect relevant correlations and trends, which significantly increases their cognitive value and interpretive capabilities for managers, allowing them to make decisions based on a more complete and processed set of data [12][7].
- **Support for diagnostic analytics** - which allows for a deeper understanding of the causes of observed phenomena through the use of cause-and-effect analysis algorithms, anomaly identification, and regression, thereby assisting decision-makers in identifying the causes of observed phenomena and taking corrective and remedial actions. In particular, ERP systems enhanced with AI components enable specialists and decision-makers to locate the sources of problems, such as disruptions in supply chains, performance declines, or increases in operating costs. Machine learning and classification models serve here as analytical tools that allow not only to understand “what happened,” but also “why it happened,” which is crucial in the context of making informed and accurate management decisions, both strategic and operational [9][7].
- **Support for predictive analytics** - which expands managers' decision-making capabilities by predicting future events and phenomena based on historical and current data. enabling the prediction of future events based on complex statistical models, neural networks, decision trees, or deep learning algorithms. Thanks to AI, ERP systems are able to forecast market demand, assess contract risk, predict production resource failures, and estimate

seasonal demand variability. These models, learning from large data sets from multiple sources (Big Data), demonstrate the ability to adapt to changing market trends and form the basis for implementing the concept of predictive business management. The predictions provided support decision-makers in resource planning, budget optimization, risk management, and preparation for changing market conditions. [13][14].

- **Prescriptive analytics** is the most developed component of the decision - making pillar, which directly supports and partially automates decision-making in the context of achieving the company's strategic goals. In this approach, the ERP system not only identifies possible courses of action, but also recommends optimal solutions from the point of view of efficiency and risk. Mechanisms based on optimization algorithms, simulations, rule-based logic, or reinforcement learning generate specific recommendations for managers, such as changing suppliers, modifying logistics processes, or adjusting the cost structure. In selected cases, it is also possible to automatically implement the recommended actions while maintaining human decision control. Thus, prescriptive analytics does not replace managers, but rather expands their decision-making capabilities by offering tools that support the selection of the most effective and rational solutions [15].

Examples of ERP systems using AI to support selected aspects of descriptive analytics include SAP Analytics Cloud (dynamic dashboards and reports), Oracle Cloud ERP (real-time automated visualizations), and IFS Cloud (reporting from IoT data, KPI analysis). In the case of ERP systems applying AI to support diagnostic analytics, examples include SAP (using AI to analyze the causes of anomalies), Oracle Cloud ERP (identification of supply chain bottlenecks, detection of financial "leakages," and analysis of problem sources), IFS.ai (for locating the causes of failures and operational costs), and Infor CloudSuite & Coleman AI (identifying issues in production and logistics). Examples of AI-enhanced ERP systems supporting predictive analytics include SAP (demand forecasts and risk analysis in S/4HANA/SAC), Oracle Cloud ERP (AutoML models for forecasting sales and inventory), IFS (predictive maintenance based on sensor data), Oracle NetSuite (NetSuite Demand Planning: real-time demand forecasting and inventory optimization), and iX ERP (predicts operational outcomes such as inventory, sales, and logistics based on historical data). ERP systems leveraging AI to support prescriptive analytics include Microsoft Dynamics 365 + Azure AI (recommendations for logistics decisions, resource management, and supply chain operations, with simulations and partial automation), SAP (Joule – recommendations for financial or supplier optimization), and IFS (IFS.ai – automatic triggering of inspections, optimized maintenance scheduling).

Within the **third area** of ERP system support by AI tools, i.e. **Interactivity, Multimedia, and Personalization of the ERP System Interface**, the following levels were distinguished:

- **Support for text and voice communication with the user** - includes support in ERP systems for processes of communicating with the user via text-based conversational interfaces, such as chatbots, which, thanks to the use of natural language processing (NLP) and user intent classification, enable interaction in a manner resembling natural human dialogue. ERP chatbots not only automate query handling, but also activate specific system functions based on semantic interpretation of commands, acting as a bridge between the user and a system with a high degree of functional complexity. In this sense, AI acts as a mediator, removing technological and language barriers to the use of ERP [16].
- **Support for multimedia dialogue and the execution of operations using voice and gestures** - at this level, ERP systems are equipped with speech recognition (ASR - Automatic Speech Recognition) and natural language interpretation (NLU - Natural Language Understanding) modules, as well as computer vision technologies that enable the analysis of gestures performed by the user. Such solutions are part of the concept of ambient intelligence, in which the system adapts to the user's behavior, rather than the other way around. This makes it possible to control ERP functionalities using voice commands (e.g., launching reports, approving forms) or gestures (e.g., when managing warehouse processes using augmented reality - AR). This type of interaction improves user comfort, facilitates mobile work, and provides better access to system functions in conditions where manual operation is limited [17].
- **Visual dialogue support and optimization of current and future tasks** - refers to intelligent assistance and optimization of user actions in real time, which is an important step towards the concept of cognitive ERP. AI tools in an ERP system analyze user activity patterns in real time, identify deviations from operating standards, suggest

corrections, suggest possible scenarios for further actions, and present critical information depending on the context of the activity. Such support is based on a combination of machine learning, predictive analytics, and business process mining techniques, enabling dynamic interface adaptation and minimizing the risk of operational errors.

- **Multimedia support for all activities and tasks** - represents a new level of interactivity in ERP systems, in which user activities are supported by advanced visual and spatial environments. The ERP system provides three-dimensional visualization of data, processes, and relationships between objects in an extended user interface, also available on multiple screens simultaneously (multidisplay technologies). Using solutions such as Extended Reality (XR), Visual Analytics, and Graph-Based Data Exploration, users can view and analyze data, organizational structures, documents, and resources in the form of interactive and spatial models. This approach is particularly effective in areas such as production planning, supply chain management, and risk analysis, where complex relationships and processes can be better understood through dynamic visualizations and simulations of various decision-making scenarios. In this context, artificial intelligence not only serves a cognitive function, but also supports the user's spatial perception, enabling full immersion in the information environment and increasing decision-making efficiency [18].

Examples of AI-supported ERP systems enabling text and voice communication with users include SAP, which offers SAP Conversational AI – chatbots integrated with the SAP Fiori interface, enabling natural language interaction. Similarly, Oracle Cloud ERP uses the Oracle Digital Assistant to support voice and text communication across its cloud suite, while IFS provides the IFS Aurena Bot, a conversational interface designed for both mobile and desktop users. In terms of multimedia dialogue support, several ERP systems have integrated advanced technologies such as speech recognition and gesture-based interfaces. For instance, SAP CoPilot allows users to issue voice commands and integrates with popular voice assistants. Oracle Voice, part of Oracle Cloud ERP, supports voice-based approval of orders and integrates with virtual assistants like Siri and Alexa. IFS has implemented AR Remote Assistance, enabling gesture-based warehouse management using augmented reality. At the level of visual dialogue support and task optimization, AI tools embedded in ERP systems help analyze user behavior in real time, identify deviations, and suggest corrective or next steps. In this context, SAP's CoPilot integrated with S/4HANA acts as an intelligent assistant that contextualizes user actions and suggests improvements (e.g., correcting invoices or workflows). Oracle Fusion ERP employs Intelligent Guidance, which offers context-aware suggestions to streamline decision-making processes. Finally, regarding multimedia support for all activities and tasks, modern ERP systems are beginning to offer fully immersive visual and spatial experiences. SAP XR Cloud provides 3D visualization of production and planning processes, enhancing comprehension of complex workflows. Oracle Cloud ERP, with tools like Oracle Visual Builder and XR integration, enables 3D data exploration and supports multidisplay dashboards. Similarly, IFS combines XR technologies with digital twins to allow interactive spatial analysis of enterprise data, especially valuable in areas like production planning and risk management.

4. A new approach to ERP systems evolution

As a result of applying AI solutions at various levels of support and within the context of the three identified areas, new types of ERP systems are emerging. In the proposed model, they are referred to as adaptive, cognitive, and fully autonomous systems, representing successive stages of ERP system maturity in terms of AI integration. However, reaching these advanced categories does not require simultaneously attaining the highest, fourth level in each of the three main areas—process automation, data analytics with decision support, and interface interactivity and personalization. Nevertheless, ERP systems should achieve at least the second or third level of maturity in each of these areas in order to develop adaptive capabilities, transition into the cognitive mode, and ultimately operate in a fully autonomous manner.

Adaptive ERP systems are modern enterprise management solutions that go beyond traditional automation and decision support functions thanks to the integration of advanced AI tools such as machine learning, deep learning, predictive analytics, and intelligent agents. Such systems can dynamically adapt their structures, operating logic, and user interface in response to changing business conditions and user behavior.

Through continuous learning based on internal and external data, adaptive ERP can optimize processes, predict events (e.g., failures, changes in demand) and automatically reconfigure workflows, increasing the flexibility and

operational resilience of the company. For example, the system can modify the sequence of activities in the order fulfillment process or reorganize the supply chain in response to current needs and priorities. An important element of such systems is interface personalization - analysis of users' work styles, preferences, and competencies allow for customization of the layout, functions, and data presented, which improves work ergonomics and speeds up decision-making, reducing the risk of errors. In addition, the use of digital twin technology enables scenario simulations and real-time assessment of the effects of decisions, supporting more accurate actions. In summary, adaptive ERP systems are a future-proof class of tools in which AI not only assists the user but also actively manages processes, enabling organizations to function as intelligent enterprises capable of adapting quickly and competing effectively in a dynamic market [19].

Cognitive ERP systems are an advanced generation of ERP systems in which artificial intelligence becomes an integral part of the decision-making and operational architecture. Going a step further than adaptive systems, they use, among other things, machine learning, natural language processing, recommendation systems, contextual intelligence, and advanced symbolic and probabilistic reasoning mechanisms. As a result, cognitive ERP not only supports the user, but also actively participates in decision-making by understanding the business context and interpreting unstructured data. Cognitive ERP systems learn on the fly from the organization's data and experience, automatically improving their recommendations and predictions. Unlike adaptive ERP, which reacts to changes, cognitive ERP can predict them and initiate actions on its own, moving from a reactive to a proactive and even partially autonomous model. An important element is natural interaction with the user - the system understands commands and questions asked in natural language, which makes it easy to use even for people outside IT departments. Practical applications of cognitive ERP include automatic production planning that takes market variables into account, financial anomaly detection, risk management, employee turnover prediction, and personalization of contact with customers and business partners. Increasingly, these systems are acting as digital advisors, supporting management in making strategic decisions based on analysis of data from multiple sources [20].

Autonomous ERP systems represent the highest level of evolution in enterprise resource management systems, where artificial intelligence is used extensively and deeply, allowing the system to make decisions independently at the operational, tactical, and even strategic levels without the need for direct human intervention. In practice, this means the transformation of ERP from a tool that supports managers' work to an autonomous, intelligent agent that manages processes in a dynamic, contextual, and adaptive manner. Autonomous ERP uses advanced AI technologies such as machine learning, deep neural networks, reinforcement learning, and complex analytical methods to analyze huge volumes of data, forecast trends, detect anomalies, and optimize processes in real time. For example, the system can independently predict demand based on seasonality, weather conditions, or market trends, and then automatically initiate orders to suppliers or modify the production schedule. Integration with the Internet of Things (IoT) allows for continuous monitoring of devices, machines, and means of transport, as well as immediate response to emergency situations, such as stopping a production line in the event of a fault or changing delivery routes in response to road conditions. As a result, the system not only responds to current data, but also predicts and prevents potential disruptions. Another important feature of autonomous ERP is its ability to self-configure and self-repair, allowing the system to automatically adapt its structures, business rules, and processes to a changing environment, e.g., by updating financial modules after a change in tax regulations without the need for user intervention. Autonomous ERP minimizes the need for direct human intervention, limiting the role of users to strategic oversight and validation of key decisions. This approach accelerates business processes, reduces errors and operating costs, and increases the resilience and scalability of the organization [21].

5. Conditions for implementing artificial intelligence integration with ERP systems

The implementation of technological innovations (such as AI) and their integration with ERP systems is a complex process requiring a multi-threaded and comprehensive approach. Scientific research and business practice indicate that the successful implementation of AI in ERP requires consideration and balancing of technical capabilities with organizational readiness. In this context, attention should be paid to [22]:

- project financing and resources;
- IT infrastructure;

- data quality and standardization of internal processes;
- competencies and organizational culture;
- external environment.

Every IT project requires the commitment of adequate financial resources. Development projects in the ERP area involve relatively high costs. Integration with AI is an additional challenge and requires even greater resources (e.g., for investments in hardware, software, specialists, or licenses). Insufficient budget is often a major barrier in the early stages of expansion. Organizations must therefore ensure long-term funding and management support for AI projects to avoid project interruption due to exhaustion of funds.

Equally important is the technological maturity of the company, including the state of its IT infrastructure. ERP systems and architecture must be ready for AI integration - this includes cloud readiness, API availability, and infrastructure scalability for real-time processing of large volumes of data. If existing systems are outdated or highly specialized, integration may require costly adjustments and adaptations, as combining AI with existing ERP can be complex and require specialized knowledge.

Furthermore, data quality and internal process standardization play a key role - AI requires high-quality, consistent data and standardized processes, otherwise algorithms may generate erroneous or useless results. For this reason, companies with higher digital and process maturity have an advantage, as structured, digitized processes and effective data management enable AI solutions to be adapted more efficiently.

The level of technological knowledge and competence within an organization determines whether AI will be properly implemented and utilized. The skills gap is a common challenge - AI integration requires a team with advanced skills in data science, machine learning, and data analysis. It is often necessary to invest in training and upskilling employees so that they can use the new tools effectively.

In addition, organizational culture and readiness for change determine the pace of AI adoption. The implementation of AI often involves a transformation of roles and processes, which can lead to resistance from employees who fear change or job losses. It is therefore crucial to implement appropriate change management, communicate the benefits of AI, and build an atmosphere of trust and openness to innovation to alleviate staff concerns. Organizations with high digital maturity typically have a culture that encourages experimentation with new technologies and continuous improvement, which facilitates the successful implementation of AI in ERP systems.

A holistic approach to the problem also includes an external factor, namely the digital and process maturity of business partners (suppliers, customers). The implementation of AI requires that partners also have the appropriate infrastructure and digital standards - otherwise, the potential of AI will not be fully exploited.

6. Conclusions

The integration of artificial intelligence with ERP systems marks a breakthrough in the evolution of enterprise resource management. The proposed multi-level AI support model which encompasses process automation, data analytics and decision support, and user interface personalization - provides a structured framework for understanding the scope and depth of AI applications within ERP environments. This model illustrates the transition from basic automation to adaptive, cognitive, and autonomous systems, highlighting the growing potential of ERP systems for independent management, adaptation, and optimization of business operations.

Artificial intelligence technologies enable ERP systems to evolve from passive data repositories into proactive agents capable of supporting, enhancing, and even independently executing business and decision-making processes. Adaptive and cognitive ERP systems give organizations the ability to respond more dynamically to market changes, while autonomous systems pave the way for entirely new operational paradigms in which systems can make strategic decisions with minimal human intervention.

However, effective integration of artificial intelligence requires more than technical implementation—it depends on organizational readiness, high-quality data, adequate IT infrastructure, financial investment, and a culture that fosters innovation. Companies that understand and are prepared for these requirements are more likely to fully harness the potential of intelligent ERP systems.

The leaders of change and the evolution of ERP systems toward adaptive, cognitive, and autonomous solutions are major business software providers such as SAP, IFS, Oracle, and Microsoft. A detailed analysis of the ERP systems offered by these providers, as well as other AI-enabled ERP solutions, will be presented in future works by the authors.

This article offers a conceptual foundation for understanding the role of artificial intelligence in the evolution of ERP systems and serves as a starting point for further empirical research. It also provides practical guidance for decision-makers and ERP solution providers seeking to navigate the complexity of AI integration and leverage its transformative potential in the digital business environment.

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